

First-Fit

Input:

```
user07@jannani:~/firstfit — /usr/bin/vim ff.c
#include<stdio.h>

int main() {
    int m, n;
    printf("Enter number of memory blocks:\n");
    scanf("%d", &m);
    int blocksizes[m], remainingsize[m];
    printf("Enter size of each block:\n");
    for (int i = 0; i < m; i++) {
        scanf("%d", &blocksizes[i]);
        remainingsize[i] = blocksizes[i];
    }
    printf("Enter the number of processes:\n");
    scanf("%d", &n);
    int processsize[n];
    printf("Enter size of each process:\n");
    for (int i = 0; i < n; i++) {
        scanf("%d", &processsize[i]);
    }
    for (int i = 0; i < n; i++) {
        int allocated = 0;
        for (int j = 0; j < m; j++) {
            if (remainingsize[j] >= processsize[i]) {
                int fragment = remainingsize[j] - processsize[i];
                printf("Process %d of size %d is allocated to block %d of size %d with fragment %d\n",
                    i + 1, processsize[i], j + 1, blocksizes[j], fragment);
                remainingsize[j] -= processsize[i];
                allocated = 1;
                break;
            }
        }
    }

    if (!allocated) {
        printf("Process %d of size %d is not allocated\n", i + 1, processsize[i]);
    }
}

return 0;
-- INSERT --
```

1.18 Top

```
user07@jannani:~/firstfit — /usr/bin/vim ff.c
    remainingsize[i] = blocksizes[i];
}
printf("Enter the number of processes:\n");
scanf("%d", &n);
int processsize[n];
printf("Enter size of each process:\n");
for (int i = 0; i < n; i++) {
    scanf("%d", &processsize[i]);
}
for (int i = 0; i < n; i++) {
    int allocated = 0;
    for (int j = 0; j < m; j++) {
        if (remainingsize[j] >= processsize[i]) {
            int fragment = remainingsize[j] - processsize[i];
            printf("Process %d of size %d is allocated to block %d of size %d with fragment %d\n",
                i + 1, processsize[i], j + 1, blocksizes[j], fragment);
            remainingsize[j] -= processsize[i];
            allocated = 1;
            break;
        }
    }

    if (!allocated) {
        printf("Process %d of size %d is not allocated\n", i + 1, processsize[i]);
    }
}

return 0;
-- INSERT --
```

1.18 Top

Output:

```
user07@jannani:~/firstfit
[user07@jannani firstfit]$ gcc ff.c
[user07@jannani firstfit]$ ./a.out
Enter number of memory blocks:
2
Enter size of each block:
150 300
Enter the number of processes:
3
Enter size of each process:
50 70 130
Process 1 of size 50 is allocated to block 1 of size 150 with fragment 100
Process 2 of size 70 is allocated to block 1 of size 150 with fragment 30
Process 3 of size 130 is allocated to block 2 of size 300 with fragment 170
[user07@jannani firstfit]$
```

Best fit:

Input:

```
user07@jannani:~ — /usr/bin/vim bestfit.c
#include<stdio.h>
int main(){
    int m,n;
    printf("Enter number of memory blocks:\n");
    scanf("%d",&m);
    int blockSize[m],remaining[m];
    printf("Enter size of each block:\n");
    for(int i=0;i<m;i++){
        scanf("%d",&blockSize[i]);
        remaining[i]=blockSize[i];
    }
    printf("Enter number of processes:\n");
    scanf("%d",&n);
    int processSize[n];
    printf("Enter size of each process:\n");
    for(int i=0;i<n;i++){
        scanf("%d",&processSize[i]);
    }
    for(int i=0;i<n;i++){
        int bestIndex=-1;
        for(int j=0;j<m;j++){
            if(remaining[j]>=processSize[i]){
                if(bestIndex == -1 || remaining[j]<remaining[bestIndex]){
                    bestIndex=j;
                }
                if(bestIndex != -1){
                    int frag=remaining[bestIndex]-processSize[i];
                    printf("Process %d of size %d is allocated to Block %d of size %d with fragment %d\n", i+1,processSize[i],bestIndex+1,blockSize[bestIndex],frag);
                    remaining[bestIndex]-=processSize[i];
                }
            }
            else{
                printf("Process %d of size %d is not allocated\n",i+1,processSize[i]);
            }
        }
    }
    return 0;
}
-- INSERT --
```

Output:

```
user07@jannani:~  
[user07@jannani ~]$ gcc bestfit.c  
[user07@jannani ~]$ ./a.out  
Enter number of memory blocks:  
2  
Enter size of each block:  
150 300  
Enter number of processes:  
3  
Enter size of each process:  
30 70 150  
Process 1 of size 30 is allocated to Block 1 of size 150 with fragement 120  
Process 1 of size 30 is allocated to Block 1 of size 150 with fragement 90  
Process 2 of size 70 is allocated to Block 1 of size 150 with fragement 20  
Process 2 of size 70 is allocated to Block 1 of size 150 with fragement -50  
Process 3 of size 150 is allocated to Block 2 of size 300 with fragement 150  
[user07@jannani ~]$
```